

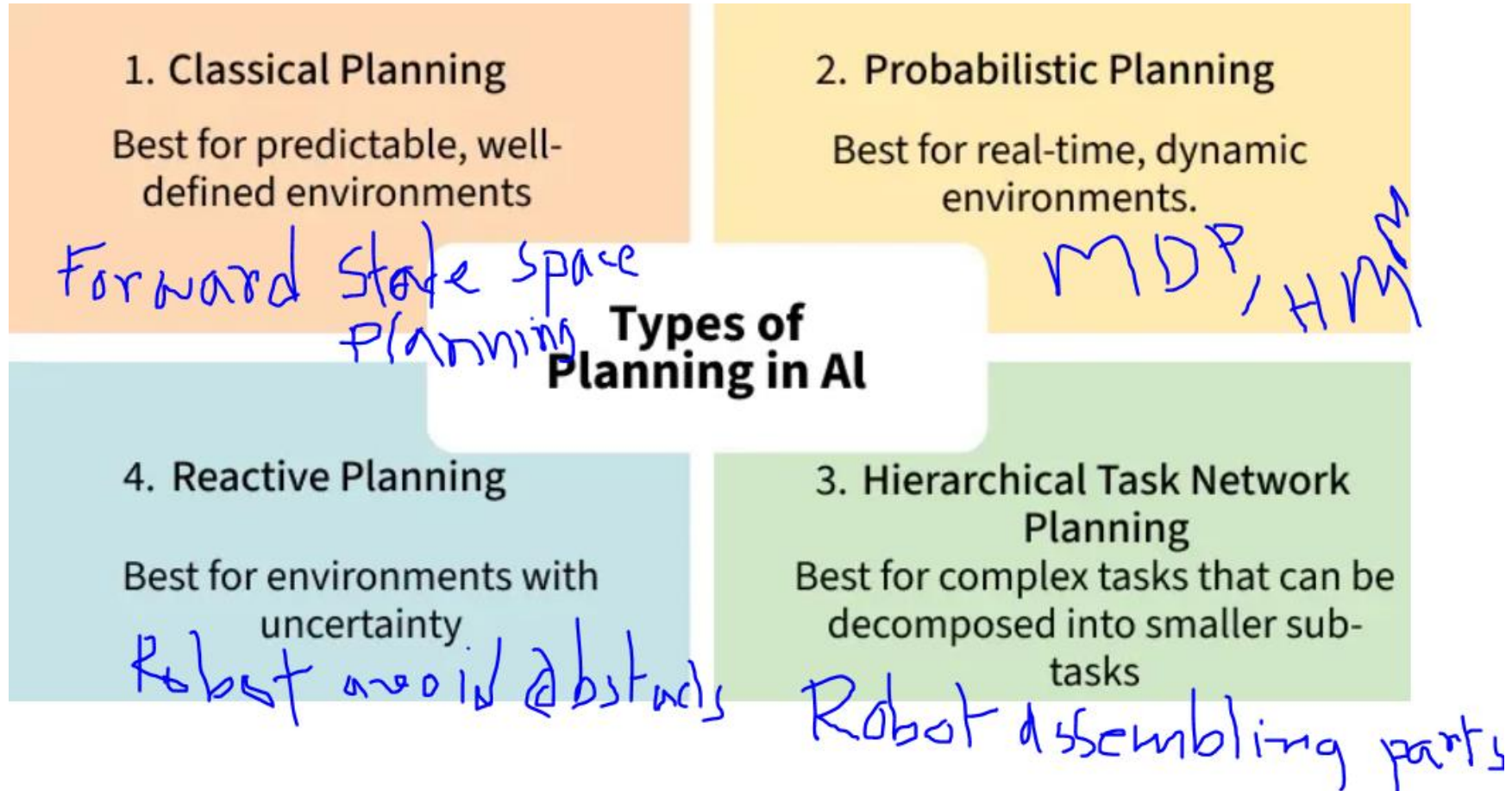
What is the Role of Planning in Artificial Intelligence?

Planning in AI refers to making a sequence of actions that leads from an initial state to a desired goal. Much like how people plan their day by outlining tasks to accomplish, AI systems use planning algorithms to break down challenges, evaluate options and determine the most effective path to success.

Key benefits of planning in AI:

- **Thinking ahead:** AI evaluates different possible outcomes and selects the best available route.
- **Adapting to change:** Planning lets AI revise its actions in response to new information or unexpected situations.
- **Autonomous action:** From robots to virtual assistants, planning enables machines to act effectively without human supervision.

Types of Planning in Artificial Intelligence (AI)



There is currently a war going on and the United States is desperate to round up all the criminals. We want to determine whether Colonel West is a criminal. Let's start with what we know.

We know that it is a crime for an American to sell weapons to hostile nations. The country Nono is an enemy of America. Furthermore, we know that Nono has some missiles, all of which were sold to it by Colonel West, who is American. (a) Represent your knowledge base using first order logic.

You can use the following function predicates: American(x), Criminal(x), Hostile(x), Missile(x), Weapon(x), Enemy(x,y), Owns(x,y), Sells(x,y,z).

(a) Represent your knowledge base using first order logic.

1. $\text{American}(x) \wedge \text{Weapon}(y) \wedge \text{Sells}(x,y,z) \wedge \text{Hostile}(z) \Rightarrow \text{Criminal}(x)$
2. $\text{Missile}(x) \Rightarrow \text{Weapon}(x)$
3. $\text{Missile}(m)$
4. $\text{Owns}(\text{nono}, m)$
5. $\text{Owns}(\text{nono}, x) \wedge \text{Missile}(x) \Rightarrow \text{Sells}(\text{west}, x, \text{nono})$
6. $\text{Enemy}(x, \text{America}) \Rightarrow \text{Hostile}(x)$
7. $\text{American}(\text{west})$
8. $\text{Enemy}(\text{nono}, \text{america})$

Forward chaining

